LESSON PLAN · SCRATCH (GR 6–8) BEGINNER: CODE FOUNDATIONS

Lesson 8: Decisions and Boolean Logic

Module 3 | 45 minutes | Selection and Boolean logic | Grades 6–8 · ages 11–14

Objective: Write programs that choose between paths with if-else, and combine conditions with and, or and not.

Success criteria

- I can use if-else and chains of decisions.
- I can combine conditions with AND, OR and NOT.
- I can test boundaries.

Key vocabulary

- **selection**: Choosing which code runs.
- **Boolean**: A value that is true or false.
- **AND / OR / NOT**: Ways to combine or flip conditions.
- **boundary**: The value where a decision changes.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Stand up if... (5 min): “stand if you have a pet AND a sibling”, then OR, then NOT.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Grade checker: learners follow the steps.
14 min	**Perform**	Code Lab: grade checker, Code Lab: ticket machine
9 min	**Probe**	Is the condition TRUE or FALSE when age = 15?, activity
4 min	**Wrap-up**	Exit ticket: When is (A AND B) true?

Unplugged warm-up

Stand up if... (5 min): “stand if you have a pet AND a sibling”, then OR, then NOT.

Teaching the idea

Selection with if-else; nested if-else implements multi-way branches (Python elif). Boolean operators AND/OR/NOT combine conditions. Boundary testing (on, just above, just below) catches off-by-one errors.

Analogy: A railway junction: the switch sends the train one way or the other, never both.

Live demo: Code the grade checker with > 90 by mistake; test 90 and reveal the bug.

Questions to ask

Ask	Listen for
How many paths run in an if-else?	Exactly one.
When is A OR B false?	Only when both are false.
Which values should you test?	On and around each boundary.

Differentiation

Support
Provide a number-line sheet with the boundaries.

Stretch
Add input validation: reject scores below 0 or above 100.

Common misconceptions

Misconception	How to fix it
AND means either	AND needs both true.
Order of the ifs doesn't matter	Check the highest range first in a chain.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
When is (A AND B) true?

Homework
Write the rules for a "should I take an umbrella?" decision using AND/OR.

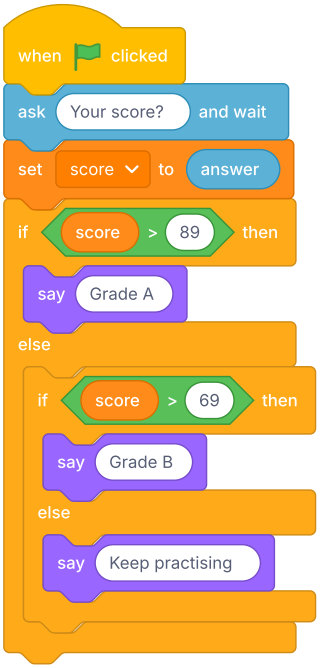
Answer key — Kitabash activities

1. Code Lab: grade checker (perform · code lab)

Goals checked: 95 → A · 75 → B · 40 → Keep practising · 90 is still an A

One solution:

Bot

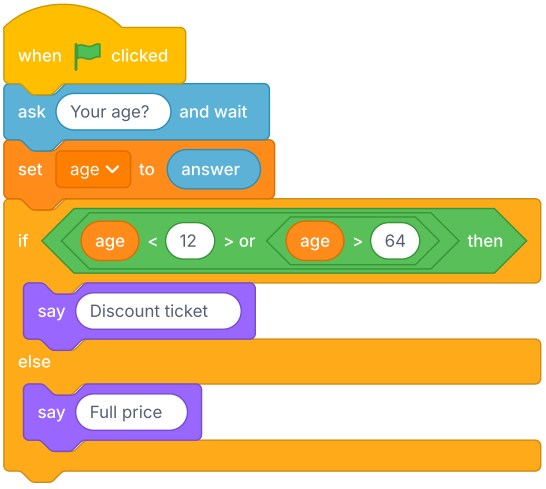


2. Code Lab: ticket machine (perform · code lab)

Goals checked: Child gets a discount · Senior gets a discount · Adult pays full price

One solution:

Kiosk



3. Is the condition TRUE or FALSE when age = 15? (probe · categorize)

True: age > 12 and age < 20, age > 10 or age > 100
False: age < 12 or age > 64, not (age = 15), age > 15

4. Explain your thinking (probe · free response)

Look for: and, test, true. **Example:** Condition: height > 120 and age > 7. Test 1: height 130, age 10 → true (can ride). Test 2: height 110, age 10 → false. Test 3 (boundary): height 120, age 8 → false, because 120 is not greater than 120.

Answer key — worksheet

1. **Key words:** selection = Choosing which code runs. · Boolean = A value that is true or false. · AND / OR / NOT = Ways to combine or flip conditions. · boundary = The value where a decision changes.
2. **Plan your code:** Goals: 95 → A; 75 → B; 40 → Keep practising; 90 is still an A.
3. **Plan your code:** Goals: Child gets a discount; Senior gets a discount; Adult pays full price.
4. **Is the condition TRUE or FALSE when age = 15?:** True: age > 12 and age < 20, age > 10 or age > 100 · False: age < 12 or age > 64, not (age = 15), age > 15
5. **Explain your thinking:** Condition: height > 120 and age > 7. Test 1: height 130, age 10 → true (can ride). Test 2: height 110, age 10 → false. Test 3 (boundary): height 120, age 8 → false, because 120 is not greater than 120.

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